Education to Eternity

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU& RTMNU

B. Tech Summer / winter 2024-25

MSE

Branch: CSE/AI SEM: -IV

Subject: Numerical Methods and Discrete Mathematics

Subject Code: CS/AI3M001 Marks: 20

Date of Exam: Time: 1 Hr.

Instructions:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to: (Listed Below CO & Mention it on question)

CO1: Describe the concept of Numerical methods, Bisection method and Newton-Raphson Method and their convergence, Gauss Elimination method and Crout's method.

CO2: Illustrate the concept of mathematical expectation, variance and standard Deviation, moment and moment generating function, characteristic function, conditional expectation, Variance and moment.

CO3: Apply the concept of joint probability and random variable, probability function Of CRV, joint probability function of DRV, joint distribution functions of DRV.

CO4: Analyse the problem by using the concept of linear dependence and independence of vector, Consistency of homogeneous and non-homogeneous equation.

CO5: Evaluate the problem based on the concept of Eigen values and Eigen vectors, Sylvester's theorem, Quadratic form, Canonical form, differentiation & integration of matrices.

CO6: Students will be able to create the new concept by using the theory of Group, group, Subgroups, Homomorphism, cosets, Lagrange's theorem and Normal subgroup.

SECTION A

Q. No.	(All Questions are compulsory).	Bloom's Level &CO	Marks
Q.1	Give an example for an algebraic and transcendental equation.	L1/CO1	1
Q.2	Define: Pivot.	L1/CO1	1
Q.3	Tell the formula for Newton- Raphson Method.	L1/CO1	1
Q.4	Illustrate the term percentage error.	L2/CO2	1

SECTION B

Q. No.	Solve answer in brief (All Questions are compulsory).		
Q.5	Interpret the terms: 1) Absolute Error 2) Relative Error	L2/CO2	2
Q.6	Illustrate the term Inherent Error.	L2/CO2	2
Q.7	Outline the matrix of the quadratic form $x_1^2 + 2x_2^2 - 7x_3^2 - 4x_1x_2 + 8x_1x_3 + 5x_2x_3$	L2/CO2	2
Q.8	Identify the interval in which root is present for the equation $x^4 - x - 10 = 0$.	L3/CO3	2

SECTION C

Q. No. **Solve any two Questions**

Q.9	Find the roots of the equation $x^4 - x - 9 = 0$ by applying Bisection method up to two iterations.	**L3/CO3**	**4**
Q.10	Inspect the roots of the equation $3x - cosx - 1 = 0$ by applying Newton-Raphson Method up to one iteration.	**L4/CO4**	**4**
Q.11	Simplify the equation by using Gauss Elimination Method. $2x + 3y + z = 13$ $x - y - 2z = -1$ $3x + y + 4z = 15$	**L4/CO4**	**4**